Lesson 03 Demo 03

Creating a React Timer App Using Vite

Objective: To set up a React project using the Vite framework, implement a timer component using functional components and React hooks, and run the application in a development environment efficiently

Tools required: Visual Studio Code and Node.js

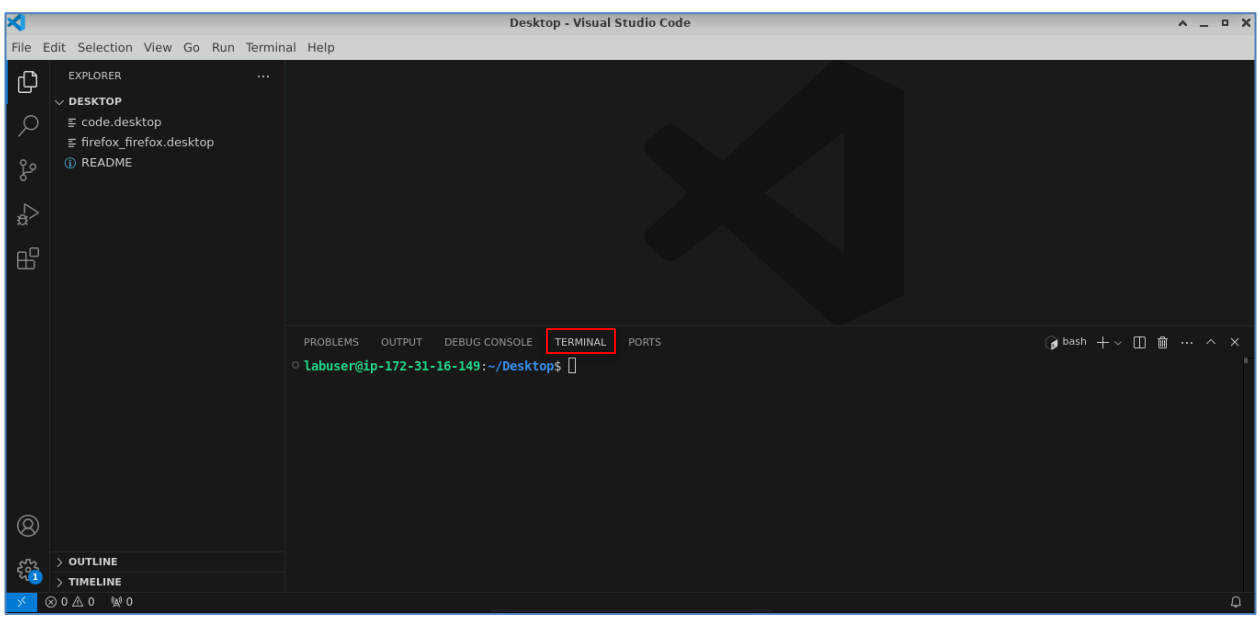
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create a React component
3. Import and use the component
4. Run and verify the application

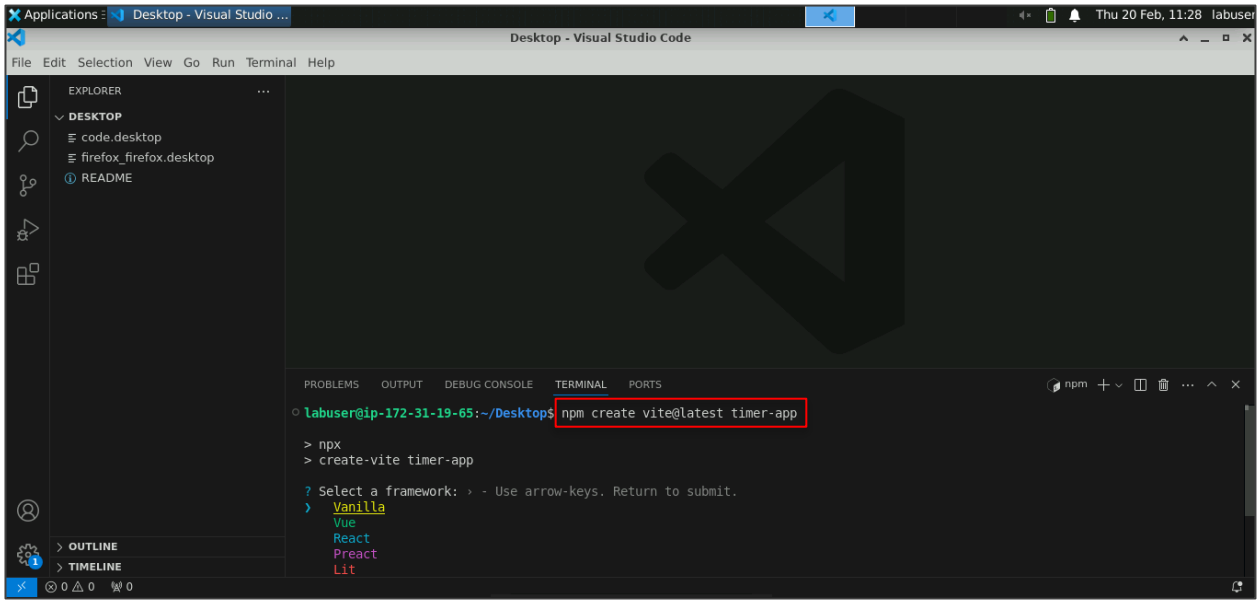
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

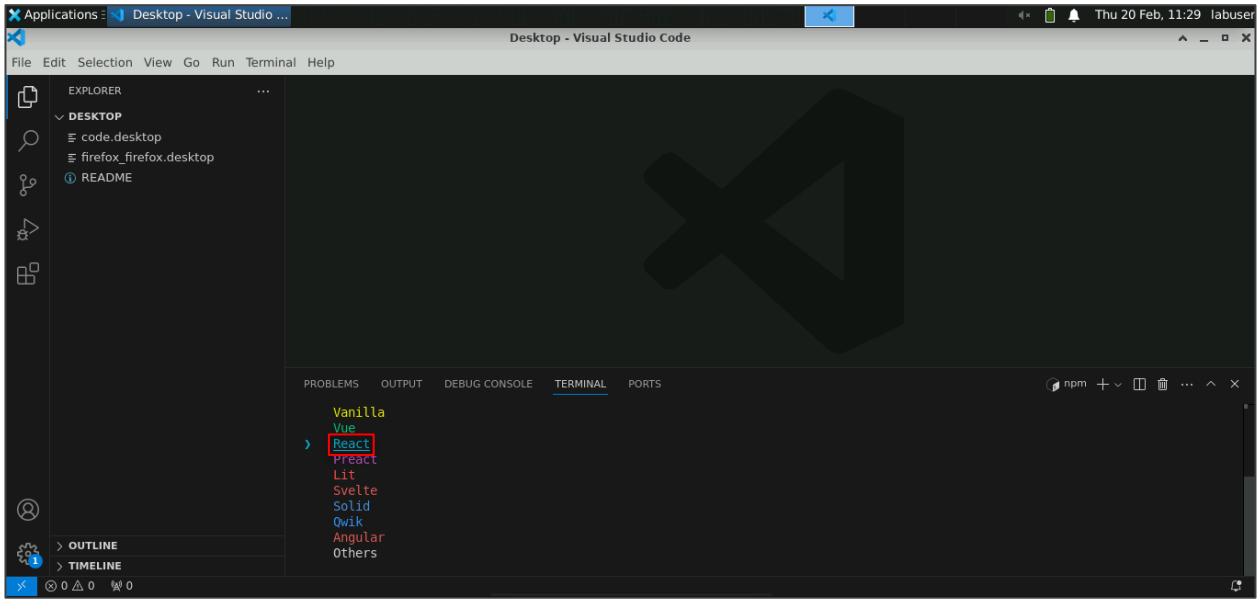


1.2 Run the following command to create a new React project using Vite:

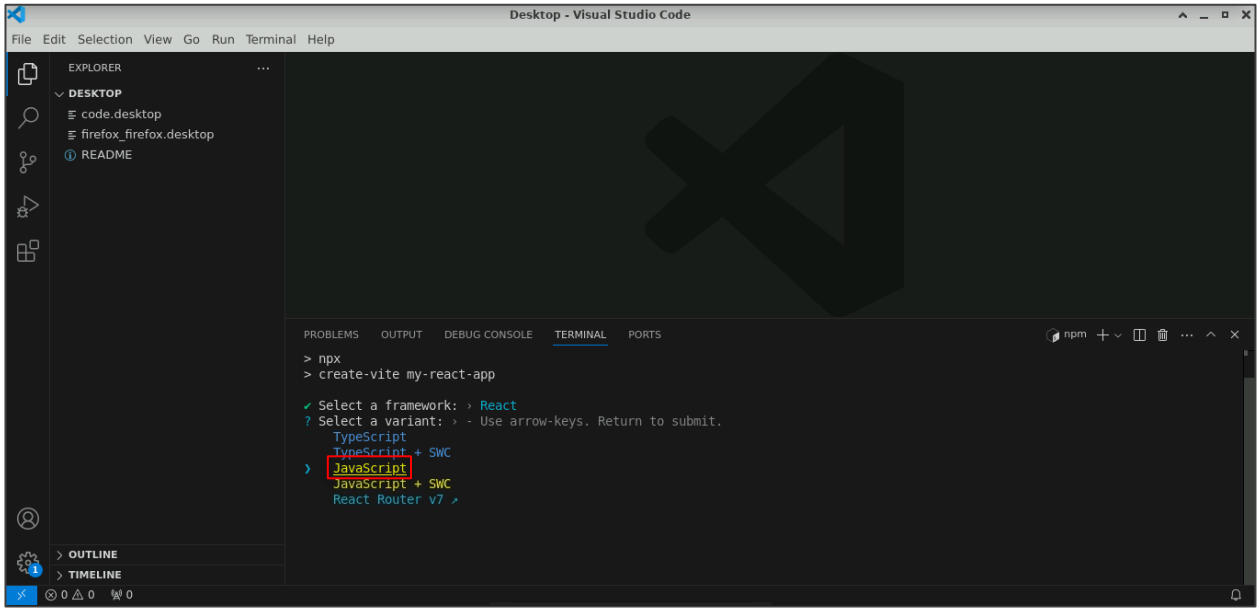
npm create vite@latest timer-app



1.3 Select **React** and press enter

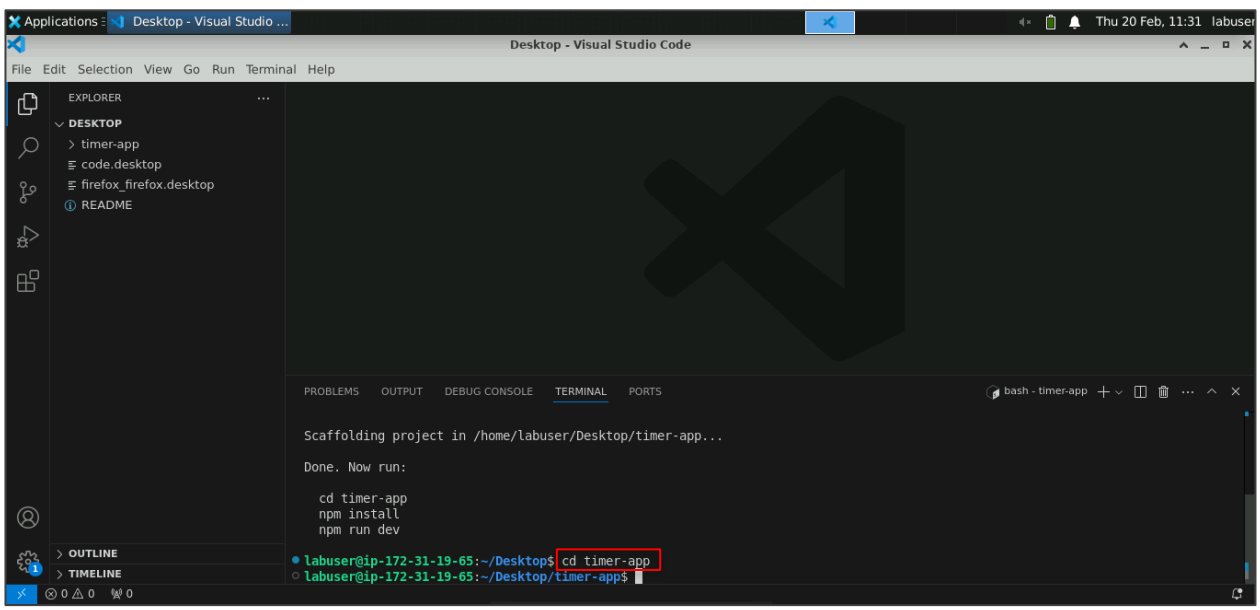


1.4 Select **JavaScript** and press enter



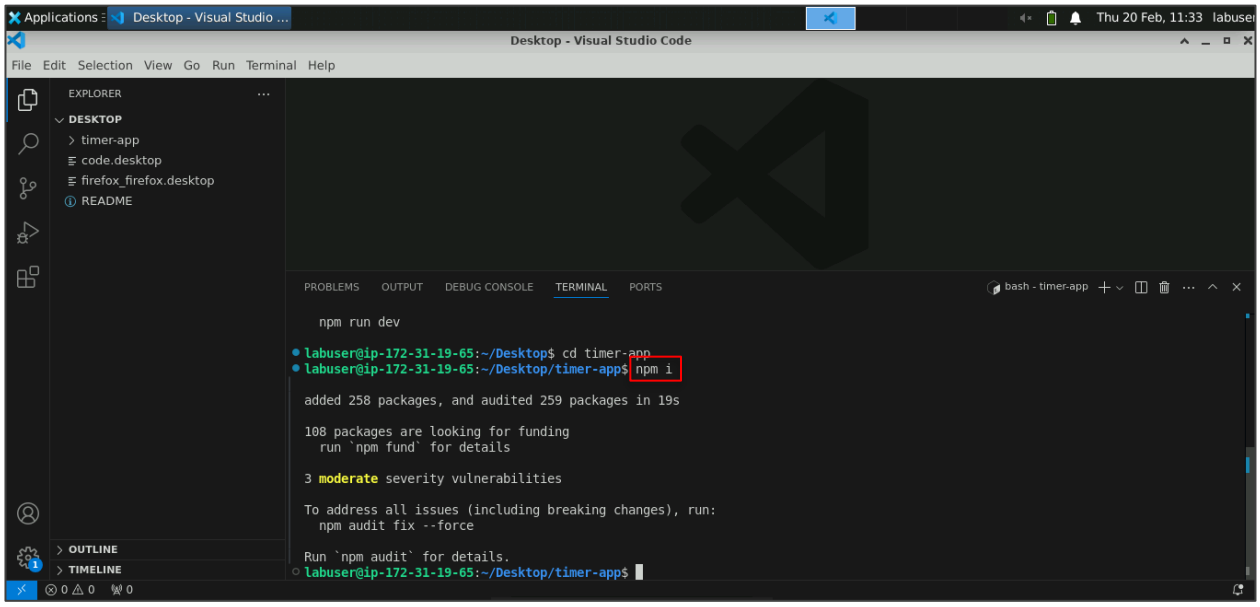
1.5 Run the following command to navigate to the project folder:

cd timer-app



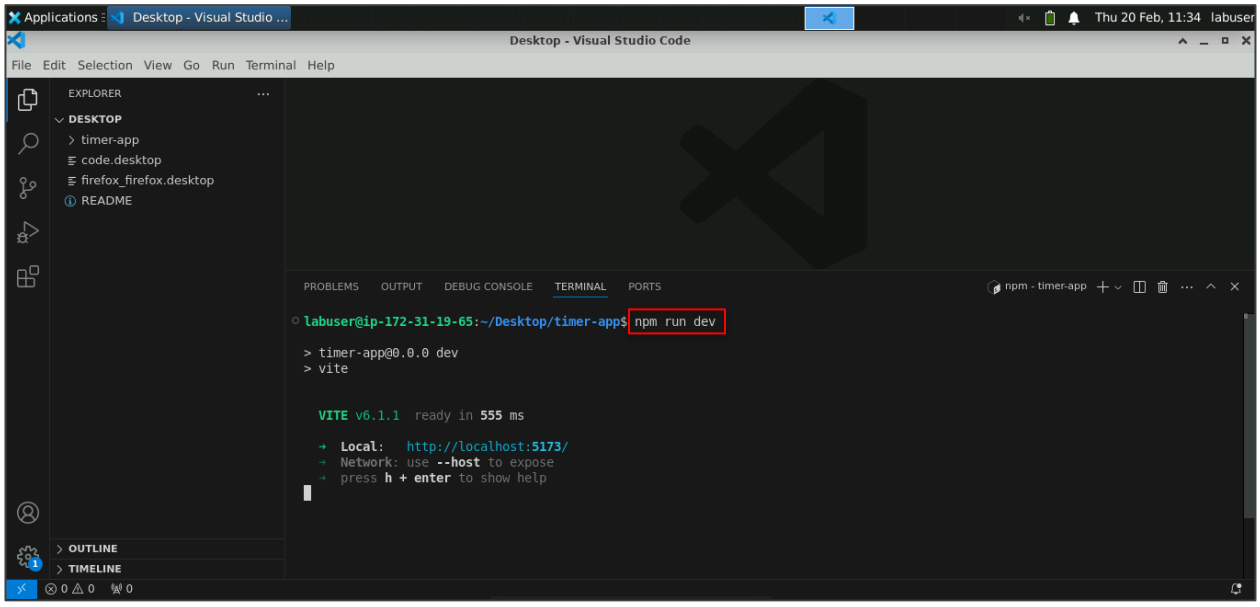
1.6 Run the following command to install dependencies:

npm i



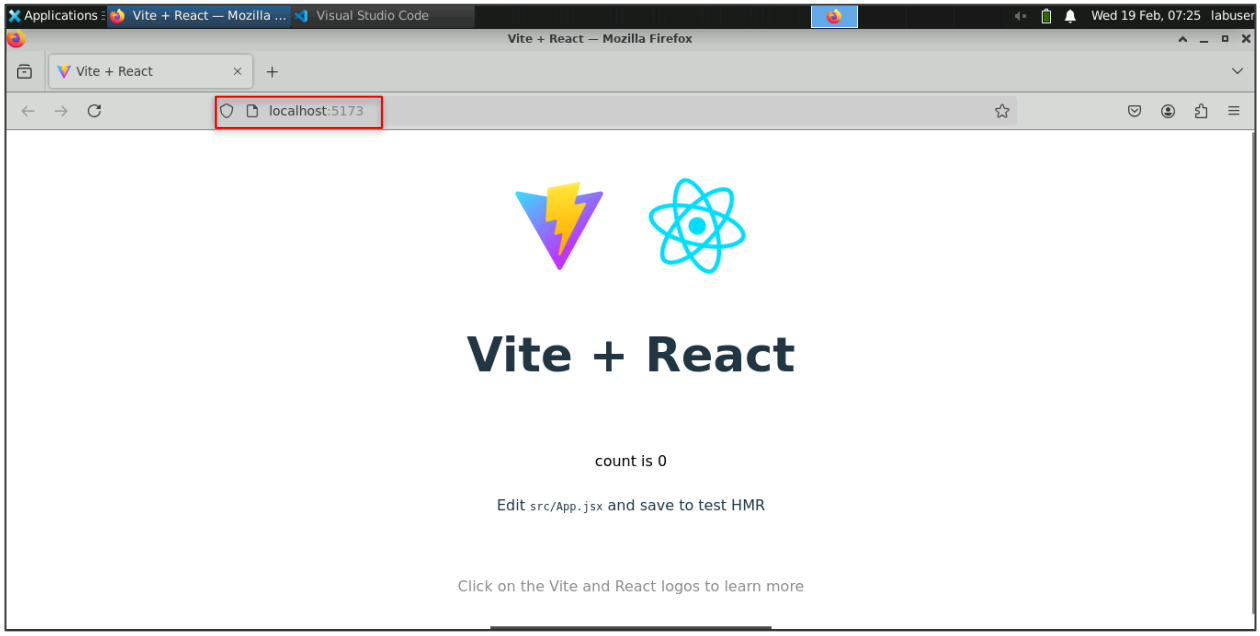
1.7 Run the following command to start the development server:

npm run dev



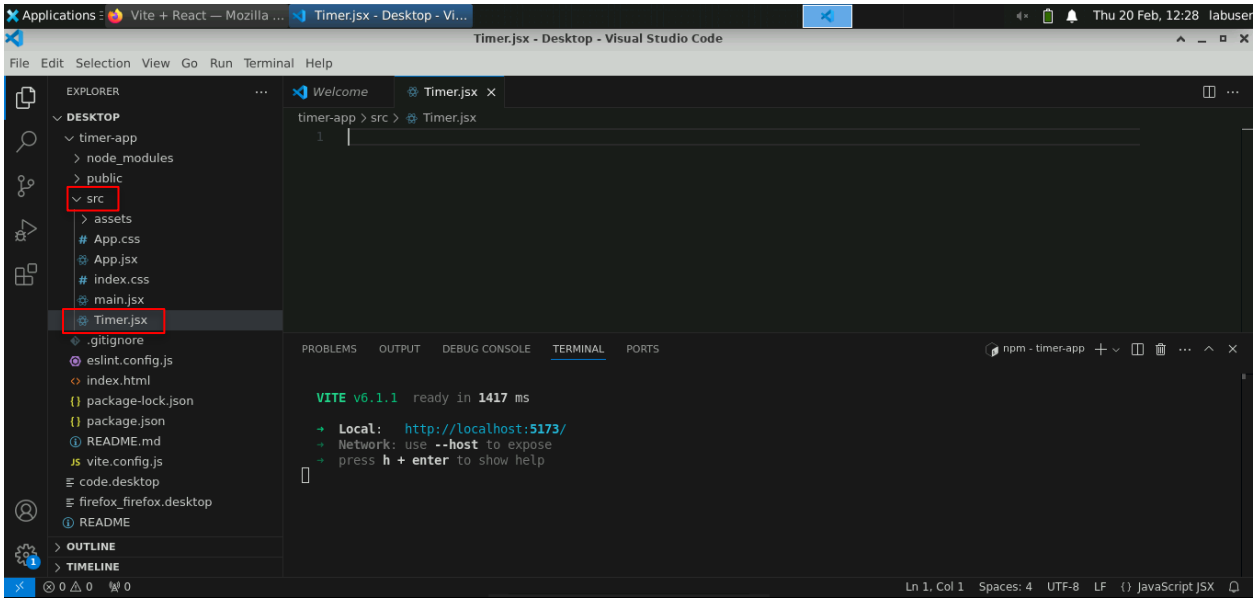
1.8 Navigate to the web browser and paste the link below:

http://localhost:5173



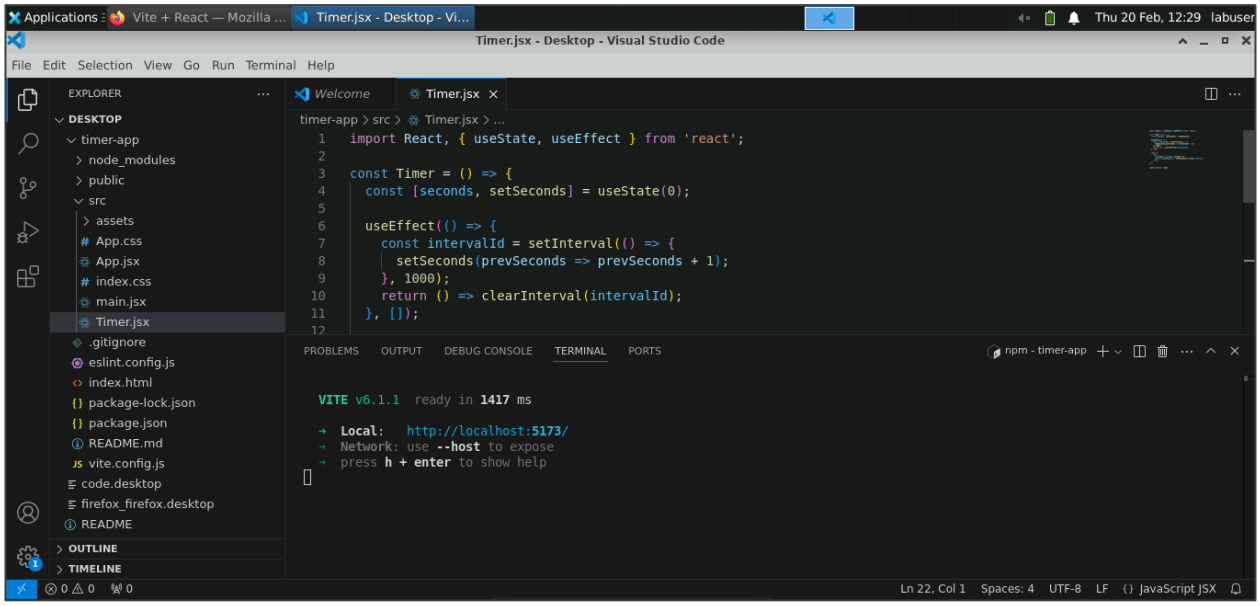
Step 2: Create a React component

2.1 Inside the **src** folder, create a new file named **Timer.jsx**



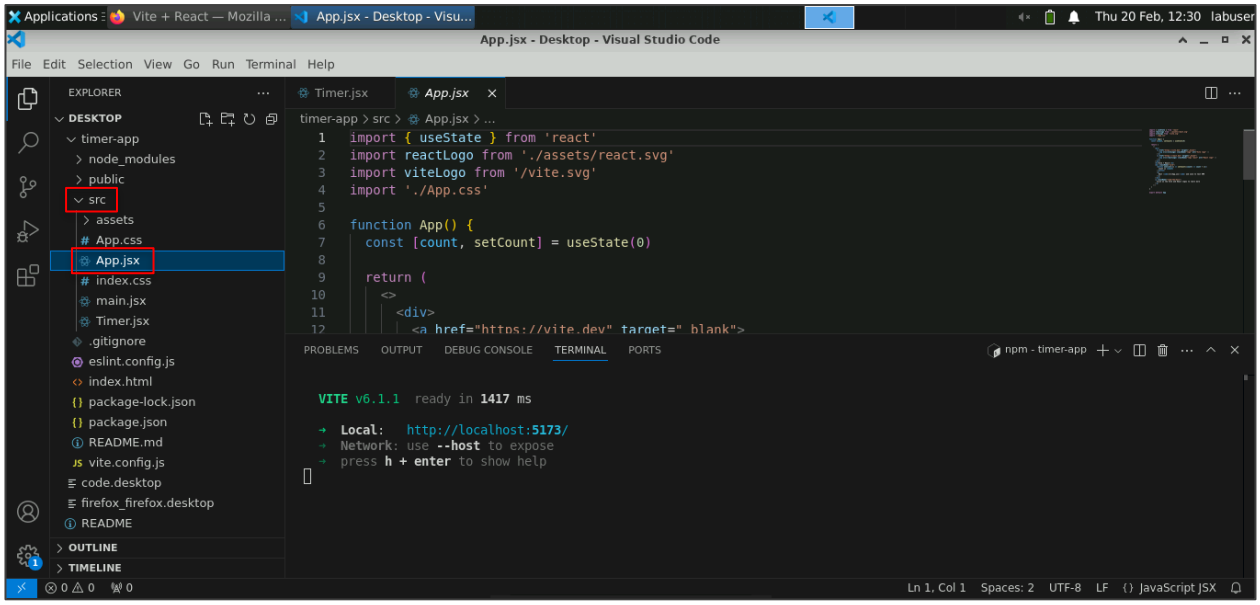
2.2 Add the following code to Timer.jsx

```
import React, { useState, useEffect } from 'react';

const Timer = () => {
  const [seconds, setSeconds] = useState(0);

  useEffect(() => {
    const intervalId = setInterval(() => {
      setSeconds(prevSeconds => prevSeconds + 1);
    }, 1000);
    return () => clearInterval(intervalId);
  }, []);

  return (
    <div>
      <h1>Timer: {seconds} seconds</h1>
      <button onClick={() => setSeconds(0)}>Reset</button>
    </div>
  );
};

export default Timer;
```



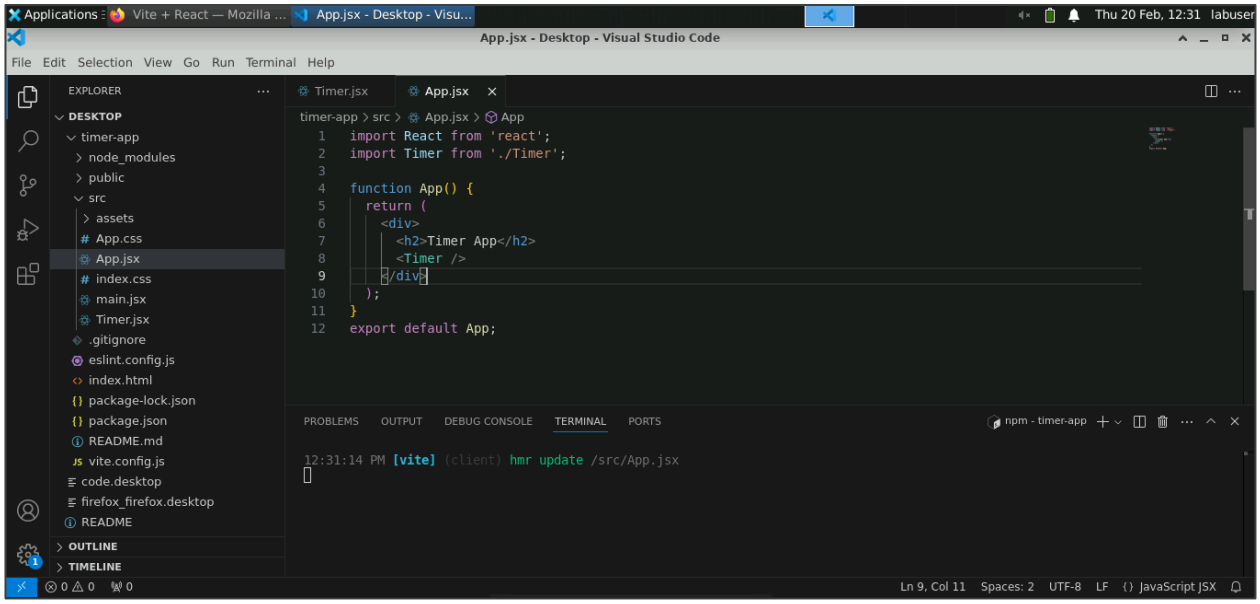
Step 3: Import and use the component

3.1 Open the **App.jsx** file inside the **src** folder



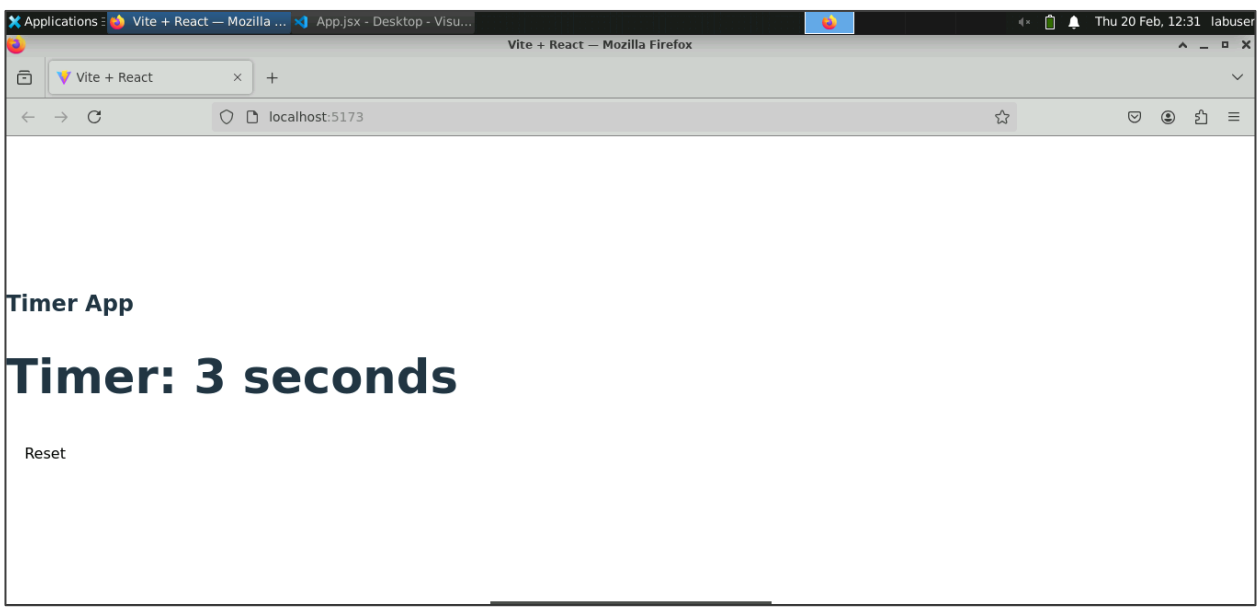
3.2 Paste the code below to include the new component in **App.jsx**:

```
import React from 'react';
import Timer from './Timer';

function App() {
  return (
    <div>
      <h2>Timer App</h2>
      <Timer />
    </div>
  );
}
export default App;
```



Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage



Note: You will see a timer that increments every second and a **Reset** button to reset it to zero.

By following the above steps, you have successfully set up a React project using Vite, created a functional timer component with React hooks, and integrated it into the application. The timer updates every second, with a reset button to reset the count.